JOURNAL OF VERBAL LEARNING AND VERBAL BEHAVIOR 7, 396–403 (1968)

Memory for the Syntactic Form of Sentences[1]

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An experiment is reported of which the purpose was to test the hypothesis that the memory for simple aspects of syntactic structure is held in the form of single, unitary, independent aspects of the memory traces for English sentences. Twenty 5-word sentences were presented to *S*s for a single trial and then *S*s were to identify the grammatical form (active, passive, negative, or question) in which each sentence occurred. If wrong, they were given a second guess. It was found that second guesses contain information. This is shown to discredit the original hypothesis. Further evidence, from *S*s' answers to a questionnaire, suggests that syntactic form is often reconstructed from other memories at the time of the retention test.

The present experiment asks whether symbols directly representing the systematic aspects of a sentence's syntactic form are found as simple independent components in the memory traces for information presented in English sentences. The method used to do so is based on research concerning the nature of recall of "attributes" of remembered materials. Bregman and Chambers (1966) have found that when *S*s encode visual material in terms of experimentally defined attributes, the remembering of each attribute in the visual displays is all-or-none. They argue that it is single descriptive features that cross an all-or-none threshold or are remembered according to some all-or-none process. Accordingly, the present experiment tries to find in the memory trace for sentences some elements whose retrieval is all-or-none and which refer specifically to syntactic features of the sentences.

Let us, then, consider memory for four types of English sentence: (a) the positive assertive active form (A), e.g., "The boy abandons the dog"; (b) the passive form (P), e.g., "The dog is abandoned by the boy"; (c) the negative form (N), e.g., "The boy does not abandon the dog"; (d) the question form (Q), e.g., "Does the boy abandon the dog?". Miller (1962), Mehler (1963), and Savin and Perchonock (1965) have found evidence to indicate that in the retention of such sentences, elements referring to the syntactic attributes of "question," "passive," and "negation" play the role of single, independently forgettable components that occupy additive amounts of "space" in the memory trace.

We are testing the hypothesis, then, as to whether simple active, passive, interrogative, and negative sentences are stored as K + A*, K + P*, K + Q*, and K + N*; here A*, P*, Q*, and N* are the names of the memory-trace components that preserve the activeness, passiveness, interrogativeness, and negativity of sentences, and K refers to the trace of all other features of the sentence, both semantic and syntactic.

If we provide K to the *S* at the time of recall, we can study the additional retrieval from memory of A*, P*, Q*, or N*. This retrieval should show all-or-none properties if A*, P*, Q*, and N* are independent components of memory traces. It is possible that A* does not exist, but is simply inferred whenever P*, N*, and Q* are absent. This in no way affects the predictions of all-or-none retention of these aspects of syntactic form.

[1] This research was financed in part by Grant no. APA-127 from the National Research Council of Canada.

METHOD

The basic experiment involves presenting four sentences, one in each of the four syntactic forms A, P, Q, and N. On the retention test *S* is given each sentence in all four syntactic forms and is to decide which sentence was in each. He is told that each form occurred once. This is a forced-choice matching procedure identical with that used by Bregman (1965). After completing the first guess, *S* is told which matches were errors and asked to guess again. If memory for each of A*, Q*, P*, or N* is all-or-none, we can infer that the second guesses would be correct at the level of chance.

Let us trace out the derivation of this prediction. If *S* fails to match the P form with the right K material on guess #1, he shows us that the original trace, K + P*, is no longer complete and that P* is missing. All that remains is K or some set of grammatical features less than K. This last statement must be true of the traces of all sentences in which there were errors on guess #1. Since only the sentences incorrectly labeled as A, P, Q, or N on guess #1 appear on guess #2, *S* has *no basis* on which to assign the forms A, P, Q, or N to these sentences on guess #2. Hence, we can predict that the number of correct matches on guess #2 will not exceed the chance level.

Subjects. Two independent replications of the experiment were conducted and are referred to as Repl. 1 and Repl. 2. In Repl. 1 all 60 *S*s were undergraduate students at McGill University. Males and females were used in approximately equal numbers. In Repl. 2, 60 female *S*s were used; 31 were student nurses and 29 were McGill undergraduates.[2]

Materials. Twenty simple, active, present-tense, affirmative, five-word sentences such as "The boy abandons the dog," were the basis of all sentences seen in the experiment. In the construction of these sentences an attempt was made to vary the most obvious semantic categories used. For example, the word "dog" was used; care was taken not to use another exemplar of the category "animal" as subject or object of a sentence. These sentences appeared in one of four syntactic forms in the experiment: A, P, Q, and N (see examples given above). The 20 basic active (A) sentences were grouped into five sets with four sentences in each set. Then, from each A sentence in each set, an N, a P, and a Q sentence were derived. Each of these sets forms a basic unit of this experiment. In the memory task, one sentence from each set appeared in the A form, one in the P form, one in the N form, and one in the Q form. Since this was true for five different sets, each *S* saw a list of 20 sentences in a mixed order. The same sentences appeared in the same forms and in the same mixed order for all *S*s. The occurrence of the four types of sentence was approximately equal in the first and last halves of the presentation order. The order of presentation was also mixed with respect to the later order of testing (which was also fixed for all *S*s). In the testing, too, the frequency of the four forms, by necessity, was balanced with respect to order of occurrence. We therefore do not expect any important order effects.

Procedure. All instructions were verbal in Repl. 1, written in Repl. 2. Instructions for the learning task were: "You will be exposed to a series of sentences, each one typed on a separate card. Exposure time is approximately 4 sec. Read each sentence carefully to yourself until you understand its meaning." The *S*s were not told that they would be tested in any way. Each card was exposed only once. Immediately after presentation, *S*s were given answer forms in which each sentence appeared in all four syntactic forms. Each set of sentences (of which one member had appeared in each form) was presented for retention testing as follows. The sentences were presented in a 4×4 matrix. The four forms of one sentence appeared in each row. There were four columns, one for each syntactic form. The *S*s were required to check off the sentences they recognized as having seen earlier. They were informed that in each set, they had seen one A, one P, one N, and one Q; i.e., they could have only one check in each column.

The *S*s were instructed to treat each set separately and to complete the recognition test as quickly as possible, although no definite time limit was specified. After completion, *E* informed *S* which sentence types had been correctly chosen. The *S*s then were allowed to use a different kind of mark to indicate a second guess among the remaining alternatives. The columns and rows in which a correct choice had been made were excluded from the second guess. The *S*s were not allowed to repeat a first-guess error. In both replications, *S*s were then told of their correct choices on the second guess.

Following completion of the retention test, *S*s of Repl. 1 were questioned orally. A written questionnaire was given to all *S*s of Repl. 2 after the retention test. In the Repl. 2 questionnaire, sets were questioned separately by means of five questions concerned with *S*s' strategies for choosing their answers on the first and second guesses. It was found that questions on all five sets were too many for *S*s. Therefore, only a sampling of two or three sets was questioned for each *S*. The sampling was done on a previously decided arbitrary basis which was designed to get examples of both good and bad performance.

[2] The authors would like to thank Miss Helene M. Lamont, Director of Nursing, and Miss Adelaide Haggart, Director of Nursing Education, of the Royal Victoria Hospital, Montreal, for helping us to obtain the volunteer *S*s used in Repl. 2.

Results

The response data consisted of the number of sentences incorrect on the first guess within each set, and the number correct on the second guess for that set. A set was not used in the data if the first guess was perfect or if it entirely constrained the second guess. The latter case occurs when there are only two sentences wrong on the first guess, since *S*s are not allowed to repeat errors. If a *S* had no admissible sets by these criteria, he was considered an inadmissible *S* in both replications. In addition, in Repl. 2, it was decided in advance to consider *S*s inadmissible if they failed to do better than chance on the *first* guess, on the grounds that there is no reason to believe that these *S*s learned anything at all. Six *S*s were eliminated for this latter reason in Repl. 2. All *S*s who were inadmissable by the above two criteria were eliminated in the course of running the experiment and experimentation was continued until 60 admissible *S*s were obtained in each replication.

First-Guess Performance. On the first guess, the mean number of correct choices over all the sets was 9.40 for the 60 admissible *S*s in Repl. 1 and was 10.05 for the 60 in Repl. 2. The expected chance value for the first guess over all five sets is five correct (one per set). The distribution of errors on the first guess suggests that there were no factors especially favoring the recall of any particular sentence type. In those sets in Repl. 1 and Repl. 2 which were used in the second-guess analysis; i.e., cases of three or four errors on guess #1, the numbers of errors were 188, 181, 192, and 189 for the A, N, P, and Q syntactic forms respectively. However, as we shall show below in the shift analysis, while *S*s did not make more errors on certain syntactic forms, they tended to make certain *types* of error.

Second-Guess Information. The *S*s contributed different numbers of sets to the second-guess data because only sets with three or four errors on the first guess are scorable. Since second-guess information can be assessed only within single sets it would be desirable to treat each set independently regardless of which *S* contributed it. There is some justification for doing this if the second-guess information in different sets from the same *S* is not correlated. Consistency within *S*s was tested as follows. The first two admissible sets from *S*s who had two or more admissible sets were used. These *S*s numbered 40 in Repl. 1 and 30 in Repl. 2. The *S*s' scores were then categorized as being above the chance expected number correct or below it, and the consistency with which *S*s stayed above or below the mean expected score in the two sets was tested. Consistency was very low in Repl. 1, χ^2 (1) = 1.02, Repl. 2, χ^2 (1) = .006, and the two combined, χ^2 (1) = .70. The lack of consistency within *S*s justifies treating each set as independent as far as second-guess information is concerned. Incidentally, this absence of within-*S* consistency in second-guess information has been a general finding in our laboratory, across a variety of experiments.

A chance frequency distribution for number correct on the second guess for cases of three or four errors on the first guess was obtained by enumeration and the data from our *S*s compared with this distribution by a chi-square

TABLE 1

Number of Correct Choices on Second Guess (Replications 1 and 2 Combined)

		Number of correct choices on second guess				
Number of errors on guess #1		0	1	2	3	4
3	Observed freq.	55	—	—	79	—
	Expected freq.	67	—	—	67	—
4	Observed freq.	12	23	37	—	16
	Expected freq.	26.05	26.05	26.05	—	9.77

test of goodness of fit (see Table 1). The obtained data differed significantly from the theoretical distribution in Repl. 1, $\chi^2(4) = 14.5$; $p < .01$, consistently in Repl. 2, $\chi^2(4) = 8.9$; $p \cong .06$, and highly reliably overall, $\chi^2(4) = 20.8$; $p < .001$. It can be seen in Table 1 that the *S*s consistently obtain higher scores on the second guess than expected by chance. There is, then, a significant amount of second-guess information.

Shift Analysis. The source of the second-guess information found in the previous section may be a subjective subcategorization of the four grammatical types on the basis of any feature of meaning that is common to two or more grammatical types. If *S* remembers such a feature, he may guess incorrectly on the first guess, and, by elimination, guess correctly on the second guess. Such subcategories would show their existence by (a) a tendency for *S*s to make errors within subcategories on their first guesses, and (b) a tendency on the second guess to shift from one member of a subcategory to another. It is, therefore, important to see whether there is any consistency among *S*s with respect to these two tendencies. Table 2 shows how the first-guess errors were distributed in those cases where there were three or four first-guess errors (i.e., in cases where a second guess was not entirely constrained). It also shows, for those same cases, the substitution of one grammatical type for another that was made on the second guess. Both the first-guess errors and the second-guess substitutions were analyzed separately for those sets where the mean number correct on the second guess was (a) above chance and (b) below chance. Data from both replications have been combined in this table. Each category of shift, for example A-P, combines changes made in both directions: from A to P and from P to A. If errors on guess #1 and substitutions on guess #2 are random, the numbers in any row of the table should vary randomly around some central value. This seems to be true where the second guess has less than the mean chance expected number of correct matches, but not in the cases where, on the second guess, more than the chance number were correct. In the latter cases, where sets showed a tendency toward second-guess information, there seemed to be two subcategories, one containing A and P, the other containing N and Q. This is shown by high entries for A-P and N-Q shifts both between the correct response and guess #1, and between the first-guess and second-guess choices (the first and third rows of Table 2; the boldfaced figures). The entries

TABLE 2

SUBCATEGORIZATION OF SENTENCE FORMS IN RESPONSE DATA

	Subcategory Type					
	A-N	A-P	A-Q	N-P	N-Q	P-Q
1. Confusion between correct form and form assigned on guess #1						
(a) Sets where guess #2 is above chance	64	**95**	69	72	**80**	65
(b) Sets where guess #2 is below chance	43	49	56	49	54	54
2. Substitution of a form on guess #2 for a form on guess #1						
(a) Sets where guess #2 is above chance	53	**110**	65	68	**95**	54
(b) Sets where guess #2 is below chance	46	51	51	44	56	57

for the six types of shift are not independent, due to the forced-choice technique, and therefore the statistical significance of these high entries cannot be evaluated by available statistical models. However, it is theoretically significant that the high values for A-P and N-Q shifts occur for only those sets in which second-guess performance is highest (above the mean expected value). This suggests that most, if not all, of the second-guess information is the result of the A-P and N-Q subcategories. (Evidence will be presented, however, that indicates that the observed N-Q subcategorization is a secondary phenomenon caused by the constraints of the forced-choice method.)

Questionnaire Data. The questionnaires were examined for evidence of why *S*s chose as they did on the first and second guesses. First of all, if we examine the reasons that *S*s gave for shifting from one grammatical type to another on the second guess, we get data that indicate that *S*s used retained information about the A-P subcategory and were able to remember it and to describe it verbally at the time of the filling out of the questionnaire. To obtain a quantitative description, we classified the shifts in sets where *S*s were questioned into (a) "Same-information" (SI) shifts, and (b) "miscellaneous" (M) shifts. Shifts were called SI if *S* said that for this shift the first and second guesses were based on the same retained partial information. All remaining shifts were classified as M. These consisted of shifts which *S* either (a) could not explain, or (b) explained by reference to information newly remembered on the second guess, or (c) explained by the constraints in the four-by-four matrix.

This categorization was done by one of us (R.S.) who then made a written set of instructions, elaborating the rules given above. On the basis of these rules, a second judge, who had no connection with the experiment, independently classified the shifts into SI and M. Of the total number of 286 shifts that were evaluated, there was disagreement on only eight cases. These eight disagreements were resolved by discussion and the final results were used in the subsequent analysis.

The first use of these data is to check the hypothesis that the second-guess information comes mainly from SI shifts. To do this a separate statistical test for second-guess information was done for the questioned sets with at least one SI shift and for the sets with no SI shifts. By themselves, the 33 sets where *S*s' protocols indicate at least one SI shift show highly significant second-guess information, χ^2 (3) = 17.29; $p < .001$. On the other hand, the 51 sets where *S*s' protocols do not indicate SI shifts show a very low and nonsignificant amount of second-guess information, χ^2 (3) = 1.69. Since second-guess information appears only where *S*s report SI shifts, the implication is that the process involved in utilizing partial memory was available to awareness, could be described reliably by *S*s, and was responsible for the second-guess information seen in this experiment.

A second use of the classification into "same-information" and "miscellaneous" shifts was to check the conclusion from the data of Table 2 that the second-guess information occurs mainly because there are two subcategories of syntactic form for *S*s, A-P and N-Q. Our data on SI and M shifts partially confirm this finding by showing us that *S*s' SI shifts were mainly concentrated in the A-P subcategory. Of the 43 SI shifts reported by our *S*s, 28 were in the A-P subcategory. Those in the remaining subcategories were as follows: A-N had 5; A-Q had 4; N-P had 0; N-Q had 5; and P-Q had 1. What of the importance of the N-Q subcategory inferred from Table 2? While N-Q shifts occurred in the data almost as often as A-P shifts, they were only reported as SI in the *S*s' protocols in five cases. The largest number of N-Q shifts is seen in sets in which some other shift (but not *that* shift) was SI. If we examine the 71 observed M shifts occurring in the same set as an SI shift, we find that 22 of these are in the N-Q subcategory. The other subcategories contained the following numbers of shifts: A-N had 9; A-P had 8; A-Q had 7; N-P had 11; and P-Q had 14. These data suggest that the large number of N-Q shifts is a secondary phenomenon caused by an interaction of the SI shift in the A-P subcategory with the constraints imposed by the forced-choice matching procedure. This can occur whenever there are four errors on guess #1. If A and P are interchanged, Q and N must also be interchanged. Substitution of A for P or P for A without the deliberate intention of complete interchange also greatly increases the likelihood of N-Q shifts relative to other types of shift. Therefore, if the protocols have any validity, there is no reason to believe that the large number of N-Q shifts represents a very strong psychological category for most *S*s.

We must now ask the question of whether the A-P subcategory is what is responsible for the observed second-guess information. We can get some evidence on this point by examining the sets where *S* could not identify *any shift* as an SI shift. As reported above, these sets contained very low and nonsignificant amounts of second-guess information. Do these sets also show a preponderance of the A-P subcategory? Apparently not. The figures are as follows: A-N had 28 shifts; A-P had 26; A-Q had 30; N-P had 21; N-Q had 39; and P Q had 28. We can conclude that since the preponderance of the A-P shift is seen only where second-guess information is high (i.e., in the SI shifts),

the A-P shift alone is accounting for most, if not all, of the second-guess information.

Phenomenological Reports. The *S*s whose protocols indicate SI shifts in the A-P subcategory expressed the nature of the A-P subcategory in the following way. They said that they knew that the event in question has *actually occurred* either by: (a) recall of the affective meaning; e.g., in the sentence "The boy abandons the dog," the *S* recalls a negative evaluative reaction which would not have occurred for the N or Q forms of this sentence; (b) recall of nonaffective aspects of meaning, such as the fact that the sentence was true; in the sentence "The opal reflects the light," the only sentence types that are true are A and P; (c) the inability of the *S* to remember either a negation or a question in the sentence; e.g., *S* remembered the sentence was "positive."

We can now turn our attention to the justifications given for choices on the *first guess* when that guess was *correct*, in order to determine what *S*s can report concerning the way in which retained information tells them what syntactic type to choose as a response. Special attention will be directed to the degree of independence of the memory traces of the syntactic features with which we are concerned from the traces of other aspects of the sentence.

(1) In our data, 33 sets were reported as being recalled "verbatim," "word for word," etc., on the first guess. One can derive very little information concerning the components of memory from this type of report.

(2) A second major category of answers implies that the recall of some internal response that *S* had made to the meaning of the sentence unambiguously determined the choice of sentence type. Questions such as "Does the Bible help the sinner?" or "Does the drug effect the cure?" were remembered as controversial questions, hence as questions; e.g., *S* remembered that "It was questioned whether drugs effect a cure." Some negatives such as "The oil does not protect the skin" were remembered on the basis of an interpretation of the sentence that was consistent with the negation; e.g., for the sentence "The oil does not protect the skin," one *S* remembered thinking that she had dry skin and therefore, that oil does not protect it.

(3) A third major category is concerned with imagery. The *S*s feel that there is a closer connection of a verb to its "surface" subject (the first noun in our sentences) than to its "deep" subject (for the distinction between "deep" and "surface" structures, see Chomsky, 1965). Hence, when there was vivid action imagery involving an idea named by a noun in a sentence, this noun was placed as the surface subject of the sentence. This more often determined the correct choice of the active form (16 cases), but sometimes the correct choice of the passive (8 cases). Connected with this in some degree is the tendency to respond to the surface subject in our sentence types as most salient, and, conversely, to place the most salient remembered noun into the surface-subject position in the sentence. When this occurred, it generally determined the choice between the active and passive form of the sentence.

(4) The last category is concerned with memories for word details or word order. For example, a remembered "ed" ending on the verb, or recall of the word "by," causes the passive to be chosen. Memory of a short "direct sentence" selects the active form. The memory of an "s" ending attached to the main verb selects the active form unambiguously. Again a *S* may report remembering that the sentence ended with a particular noun, this being sufficient to determine the choice of the passive if this noun is the "actor" in the sentence. Memory for the order of the two nouns is sufficient to select or to rule out the passive form.

In summary, it can be seen that in our experiment the information used to decide upon syntactic form resided in all sorts of memories including memories of word order, word endings, subject salience, affective response, truth value, interpretation, and action imagery. In some sense then, it seems that the grammatical form is encoded into some set of these factors and, in this manner, information that reflects the original syntactic form is preserved in these memories.

Discussion

Miller (1962) and Mehler (1963) imply that each of the syntactic contrasts between a memorized sentence and a standard A sentence is represented in the memory trace by the addition of one "footnote." Let us call this the "syntactic symbol" theory. If this syntactic symbol were the only representation of the corresponding grammatical contrast, then *S*s' retention of sentence form should have been all-or-none. The fact that this was not all-or-none shows that our *S*s used more than one aspect of their memories to decide on the syntactic form of the sentences. In the present experiment, these multiple features used by our *S*s to reconstruct grammatical form seemed to have reference to a number of *semantic* features, listed at the end of the previous section. This finding requires a theory of sentence recall in which a distinction is drawn between the *transmission code* and the *semantic message*. When the medium of communication is some particular language, the properties of

the transmission code are the forms of that language (e.g., the A, P, N, or Q forms of English). These forms provide information about (a) properties of the semantic message (e.g., what ideas are selected and how they are connected) and (b) information to the decoding processes as to what rules to use to decode the message.

Our present findings suggest that there is no reason to believe that after the properties of the transmission code have done their job and the semantic message has been decoded, the properties of the transmission code need be reflected in any explicit or uniquely decodable manner in the properties of the semantic message. It is further hypothesized that it is to the properties of the semantic message that long-term memory-code elements refer under most types of natural listening or reading conditions, and not to the properties of the transmission code. What this means is that under most conditions, such features as passivization are stored implicitly in the trace, in the form of meanings implied by the use of that feature, rather than explicitly by the use of directly relevant code elements.

We believe that this theory makes the most sense out of our data. The existence of the A-P subcategory may be explained by the hypothesis that the primary response to a message concerns whether it expresses a fact and, if so, whether the fact is true or false, and what it implies for belief, attitude, or action. Since passivization has little implication for these major concerns about messages, the properties of the message most likely to be stored do not reflect the passivization directly, and the memory trace obtained from passive sentences will differ only in minor ways from the memory trace obtained from the corresponding active sentences.

Another point to recall is that our data strongly suggest that in the case where a *S* did not get a negative or question correct on the first guess, there was no evidence that he could do better than chance on the second guess. The only case where guess #2 is correct with a better-than-chance frequency is where the original sentence was passive or active. That is, when the *passive* feature is forgotten, the remaining trace has something particularly similar to the trace of an active sentence, but the trace of a sentence with a forgotten *negative* feature has not. This is consistent with the theory of the semantic coding of sentences in memory.

We must, however, modify this theory somewhat to explain why Mehler (1963) and Savin and Perchonock (1965) got results favoring the interpretation that there *are* code elements in the memory trace directly referring to the syntactic type of the sentence. We can think of two possible explanations: (a) We may postulate that the syntactic features of a sentence must be identified and information about them preserved *briefly* in order for the sentence to be decoded (perceived). However, we might assume that information directly related to sentence form is generally dropped after doing its job. This might explain the difference between Savin and Perchonock's findings and our own as the result of differences in the time interval over which memory was tested. The syntactic symbols might still have been present in the memory trace over the 6- to 10-sec recall intervals of Savin and Perchonock, but have disappeared in our recall interval (a period on the order of a few minutes). (b) We might postulate that a rote-memorization set alters the nature of the encoding. Savin and Perchonock (1965) instructed their *S*s to recall verbatim. Mehler (1963) does not report his instructions, but his procedure of repeated presentations and repeated written recalls undoubtedly fostered a rote-learning set in *S*s. The present experiment tried not to do so. Our *S*s were not instructed to memorize the sentences, but simply to understand their meaning. Furthermore, since we gave only one exposure trial, *S*s could not know what the nature of the testing would be. If a rote-memorization set is required before *S*s will encode the syntactic features explicitly, this might explain the rareness of the report of

specifically syntactic memories in our experiment. It is not that we believe that *S*s *cannot* encode transmission code properties of messages; we believe that they usually *do not*.

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(Received January 10, 1967)